

Chapter Two

Intelligent Agent

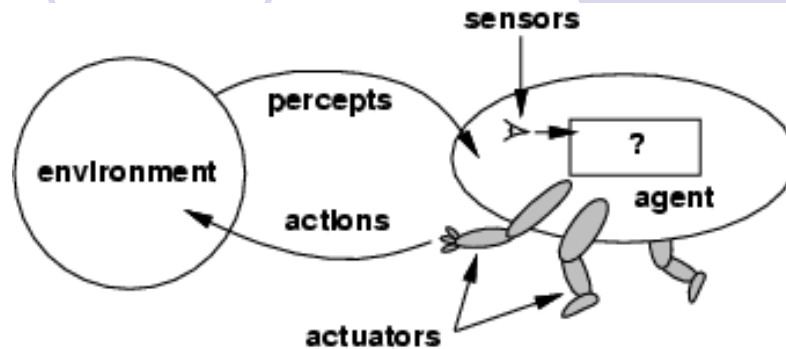
- **Objective:**

- Define what an agent is in general
- Gives ideas about agent, agent function, agent program and architecture, environment, percept, sensor, actuator (effectors),
- Gives idea on how agent should act
- How to measure agent success
- Rational agent, autonomous agent
- Types of environment
- Types of agent

- What is an agent?

- Is any thing that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon the environment through the **effectors**
- Human as an agent has eyes, ears, and other organs for sensors; and hands, legs, mouth, and other organs as effectors
- Robots as an agent has camera, sound recorder, infrared range finder for sensors; and various motors for effectors.

Agents and environments



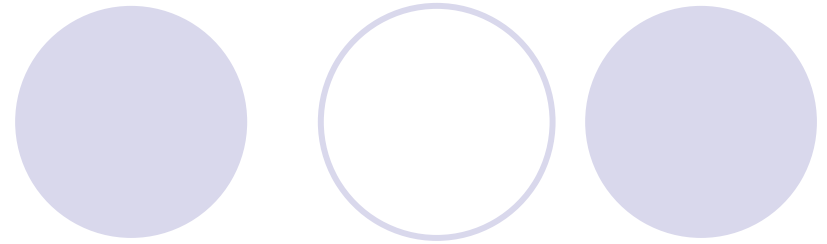
- The **agent function** maps from percept histories to actions:

$$[f: P^* \rightarrow A]$$

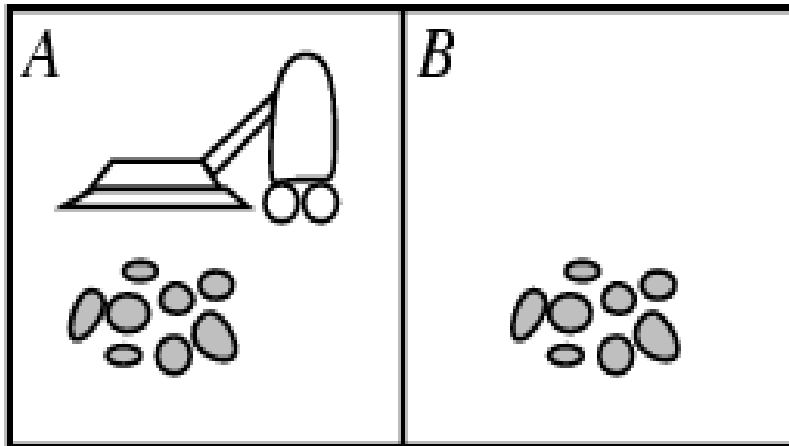
- The **agent program** runs on the physical **architecture** to produce f
- $\text{agent} = \text{architecture} + \text{program}$

Ideal Example of Agent

❖ Vacuum-cleaner world



- **Percepts:** location and contents
 - e.g., [A, Dirty]
- **Actions:** *Left, Right, Suck*



How should Agents act?

- Agents may be rational or human like
- We have seen how human act or think is difficult to understand since due to the complex structure of human intelligence.
- Our agent should be designed from rationality view that act rationally
- A rational agent is an agent that does the right thing for the perceived data from the environment
- **What is right** is an ambiguous concept but we can consider the right thing as the one that makes the agent **more successful**.
- Success is also measured by using performance measure
- Question: **how** and **when** do you measure success in performance?

- **Performance measure (how?)**

- **Subjective measure using the agent**

- How happy is the agent at the end of the action
- Agent should answer based on his opinion
- Some agents are unable to answer, some are delude themselves, some over estimate and some under estimate their success
- Therefore, subjective measure is not a better way.
- Objective Measure imposed by some authority is an alternative

- **Objective Measure**

- Needs standard to measure success
- Provides quantitative value of success measure of agent
- Involves factors that affect performance and weight to each factors
- E.g., performance measure of a vacuum-cleaner agent could be
 - amount of dirt cleaned up,
 - amount of time taken,
 - amount of electricity consumed,
 - amount of noise generated, etc.
- The time to measure performance is also important for success.
- It may include knowing starting time, finishing time, duration of job, etc

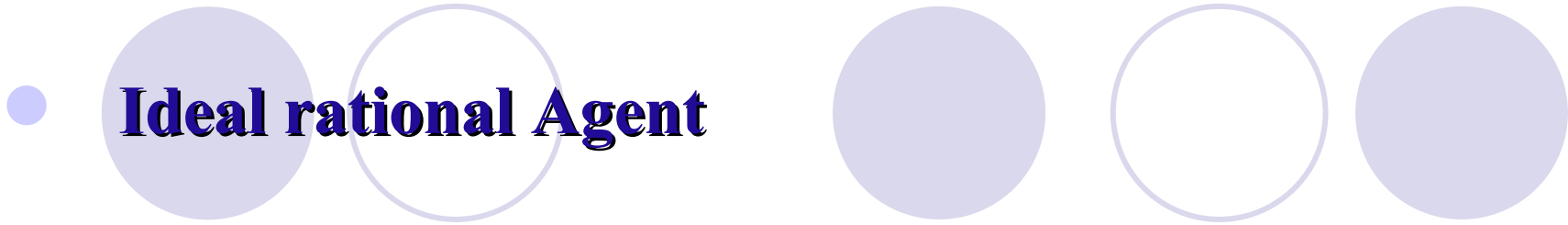
- **Omniscience versus Rational Agent**
- Omniscience agent is distinct from Rational agent
- Omniscience agent is an agent that knows the actual outcome of its action. However, rational agent is an agent that tries to achieve more success from its decision
- Rational agent could make a mistake because of unpredictable factors at the time of making decision.
- Omniscient agent that act and think rationally never make a mistake
- Omniscient agent is an ideal agent in real world
- Agents can perform actions in order to modify future percepts so as to obtain useful information (information gathering, exploration)



● Factors to measure rationality of agents

1. Percept sequence perceived so far (do we have the entire history of how the world evolve or not)
2. The set of actions that the agent can perform (agents designed to do the same job with different action set will have different performance)
3. Performance measures (is it subjective or objective? What are the factors and their weights)
4. The agent knowledge about the environment (what kind of sensor does the agent have? Does the agent knows every thing about the environment or not)

● This leads to the concept of **Ideal Rational Agent**



● Ideal rational Agent

- For each possible **percept sequence**, an ideal rational agent should do **what every action** is expected to maximize its **performance measure**, on the basis of the evidence provided by the **percept sequence** and what ever **built-in knowledge** the agent has.
- Ideal rational agent implementation require perfection
- In real situation such agent is difficult to achieve
- Why car accident happened? Because drivers are not perfect agent

Autonomy

- An agent is **autonomous** if its behavior is determined by its own experience (with ability to learn and adapt)
- Agent that **lacks autonomous**, if its actions are based completely on built-in knowledge
- Example: student grade decider agent:
 - Knowledge base given: rules for converting numeric grade to letter grade
 - Case 1: agent always follows the rule (lacks autonomous)
 - Case 2: agent that modify the rules by learning exceptions from the knowledge base as well as grade distribution.

Structure of Intelligent Agent

- Structure of AI Agent refers to the design of **intelligent agent program** (function that implement agent mapping from percept to actions) that will run on some sort of **computing device** called **architecture**
- This course focus on intelligent agent program function theory, design and implementation
- Design of intelligent agent needs prior knowledge of
 - **Performance measure** or **Goal** the agent supposed to achieve,
 - On what kind of **Environment** it operates
 - What kind of **Actuators** it has (what are the possible **Actions**),
 - What kind of **Sensors** the agent has (what are the possible **Percepts**)
- **Performance measure** → **Environment** → **Actuators** → **Sensors** are abbreviated as PEAS
- **Percepts** → **Actions** → **Goal** → **Environment** are abbreviated as PAGE

Examples of agents structure and sample PEAS

- **Agent: automated taxi driver:**
 - **Environment:** Roads, traffic, pedestrians, customers
 - **Sensors:** Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard
 - **Actuators:** Steering wheel, accelerator, brake, signal, horn
 - **Performance measure:** Safe, fast, legal, comfortable trip, maximize profits
- **Agent: Medical diagnosis system**
 - **Environment:** Patient, hospital, physician, nurses, ...
 - **Sensors:** Keyboard (**percept** can be symptoms, findings, patient's answers)
 - **Actuators:** Screen display (**action** can be questions, tests, diagnoses, treatments, referrals)
 - **Performance measure:** Healthy patient, minimize costs, lawsuits

Examples of agents structure and sample PEAS

- **Agent:** Interactive English tutor
 - **Environment:** Set of students
 - **Sensors:** Keyboard (typed words)
 - **Actuators:** Screen display (exercises, suggestions, corrections)
 - **Performance measure:** Maximize student's score on test
- **Agent:** Satellite image analysis system
 - **Environment:** Images from orbiting
 - **Sensors:** Pixels of varying intensity, color
 - **Actuators:** print categorization of scene
 - **Performance measure:** Correct categorization
- **Agent:** Part picking robot
 - **Environment:** Conveyor belt with parts
 - **Sensors:** pixels of varying intensity
 - **Actuators:** pickup parts and sort into bins
 - **Performance measure:** place parts in correct bins

Agent programs

- An agent is completely specified by the agent function that maps percept sequences into actions
- Aim: find a way to implement the rational agent function concisely
- Skeleton of the Agent

***FUNCTION SKELETON-AGENT** (percept) returns action*
static memory, the agent's memory of the world
memory $\leftarrow$ *UPDATE-MEMORY* (memory, percept)
action $\leftarrow$ *CHOOSE-BEST-ACTION* (memory)
memory $\leftarrow$ *UPDATE-MEMORY* (memory, action)
RETURN action

Note:

1. the function gets only a single percept at a time
2. The goal or performance measure is not part of the skeleton

Table-lookup agent

- Table look up agent store all the percept sequences –action pair into the table
- For each percept, this type of agent will search for the percept entry and return the corresponding actions.
- Table look up couldn't be the right option to implement successful agent
- Why?
- Drawbacks:
 - Huge table
 - Take a long time to build the table
 - No autonomy
 - Even with learning, need a long time to learn the table entries

Agent types

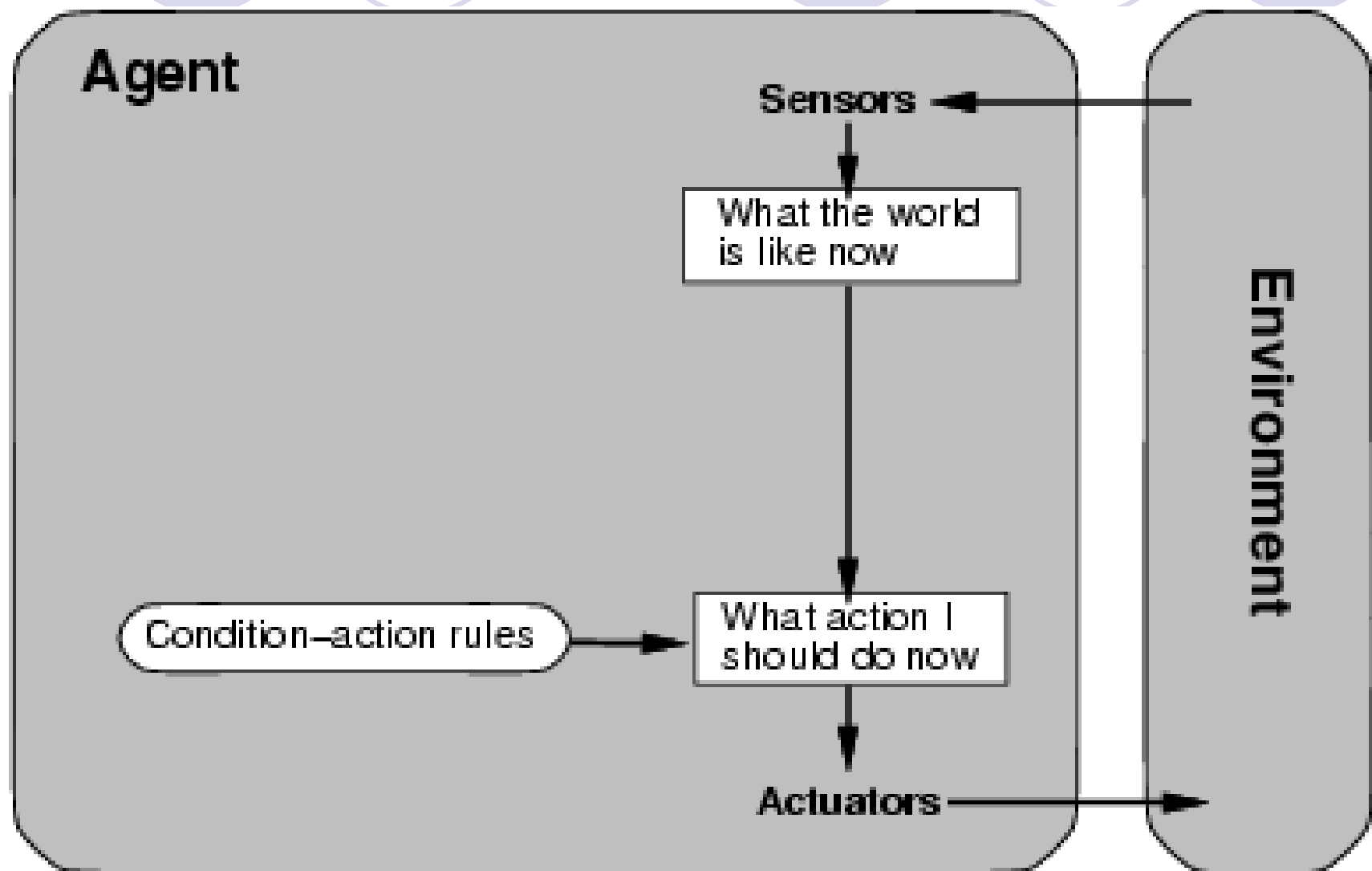
- Based on **memory of the agent**, and the way the **agent takes action** we can divide agents into five basic types:
- These are (according to their increasing order of generality) :
 1. Simple reflex agents
 2. Model-based reflex agents
 3. Goal-based agents
 4. Utility-based agents
 5. Learning agent

Each will be discussed soon with models.

Notation of model:

- **Rectangles**: used to represent the current internal state of the agent decision process
- **Ovals**: used to represent the background information used in the process

Simple reflex agents



Simple reflex agent function prototype

- Function `SIMPLE_REFLEX_AGENT(percept)` return action
static : rules, a set of condition –action-rule

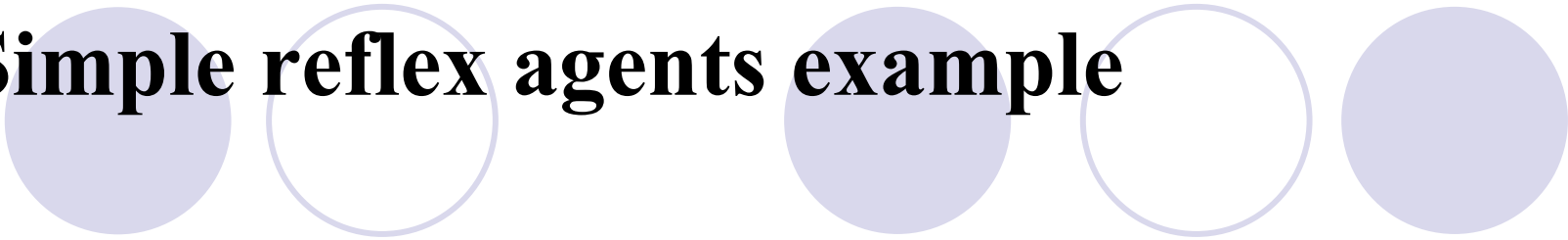
state $\leftarrow$ *INTERPRET-INPUT*(*percept*)

rule $\leftarrow$ *RULE-MATCH*(*state*, *rules*)

Action $\leftarrow$ *RULE-ACTION*[*rule*]

Return action

Simple reflex agents example



Consider an artificial robot that stand at the center of Meskel square (environment)

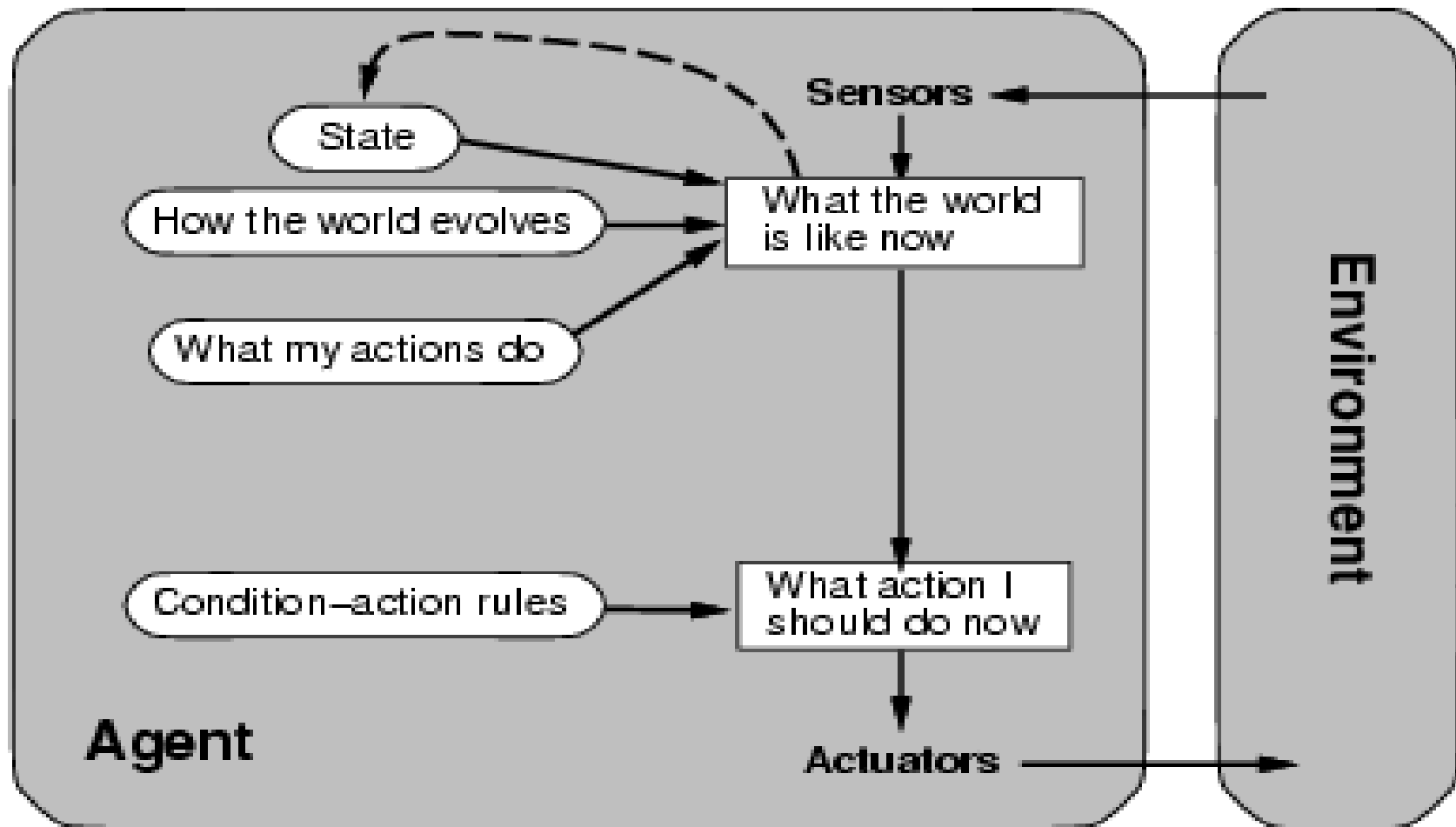
Agent has camera and microphone (sensor)

if the agent perceive a sound of very high frequency (say above 20Khz) the it start to fly up to the sky as much as possible

if the agent perceive an image which looks like a car it run away in the forward direction

Otherwise it just turn in any direction randomly

Model-based reflex agents (also called a reflex agent with internal state)

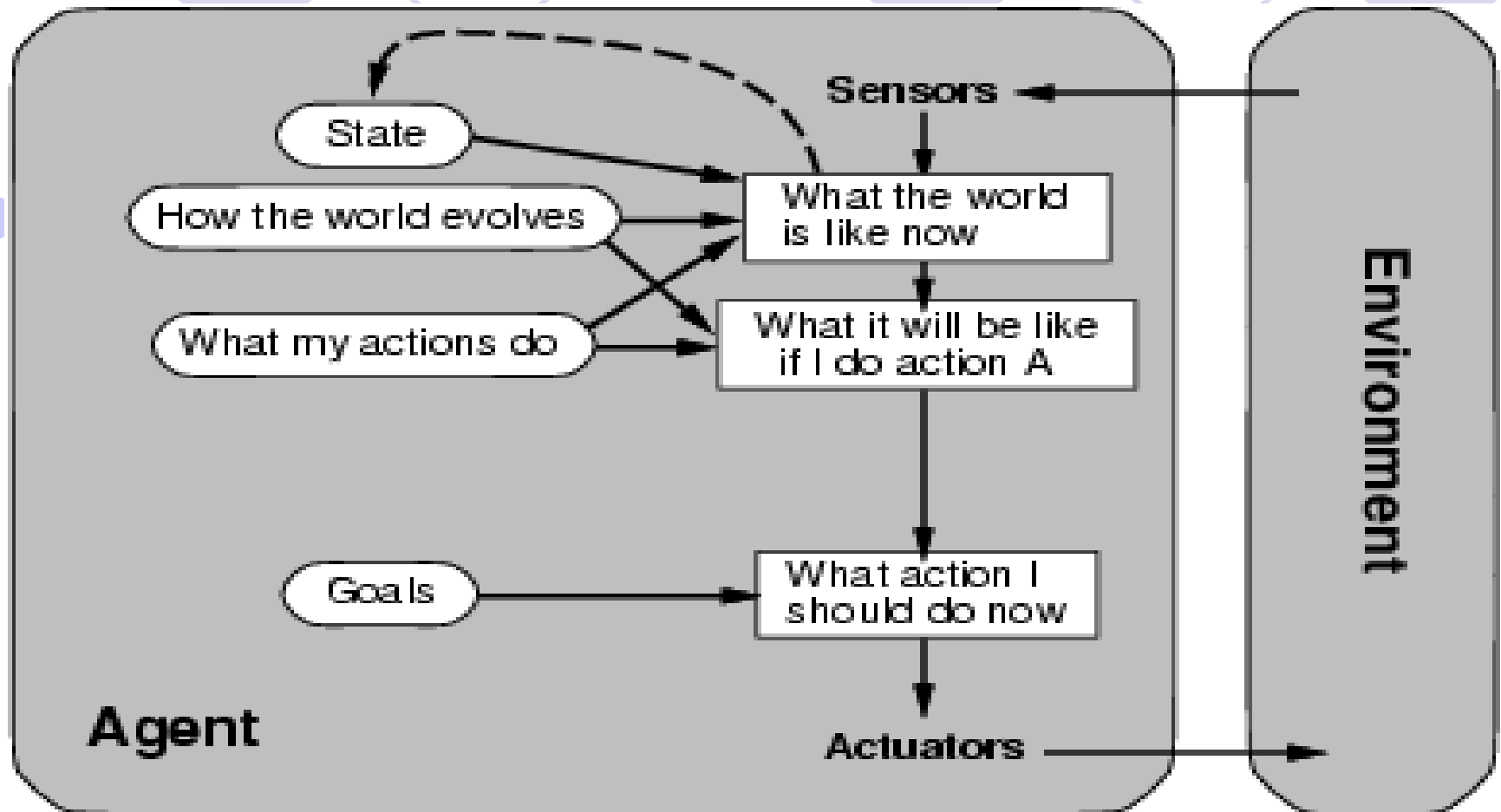


Model-based reflex agents

- Function `MODEL_BASED_AGENT(PERCEPT)` return action
`static state`, a description of the current world state
`rules`, a set of condition action rules

```
state ← UPDATE_STATE(state, percept)  
rule ← RULE_MATCH(state, rues)  
action ← RULE_ACTION[rule]  
state ← UPDATE_STATE(state, action)  
return action
```

Goal-based agents



Goal-based agents structure

- Function `GOAL_BASED_AGENT(PERCEPT)` return action
static state, a description of the current world state

goal, a description of the goal to achieve may be in terms of state

state $\leftarrow$ `UPDATE_STATE(state, percept)`

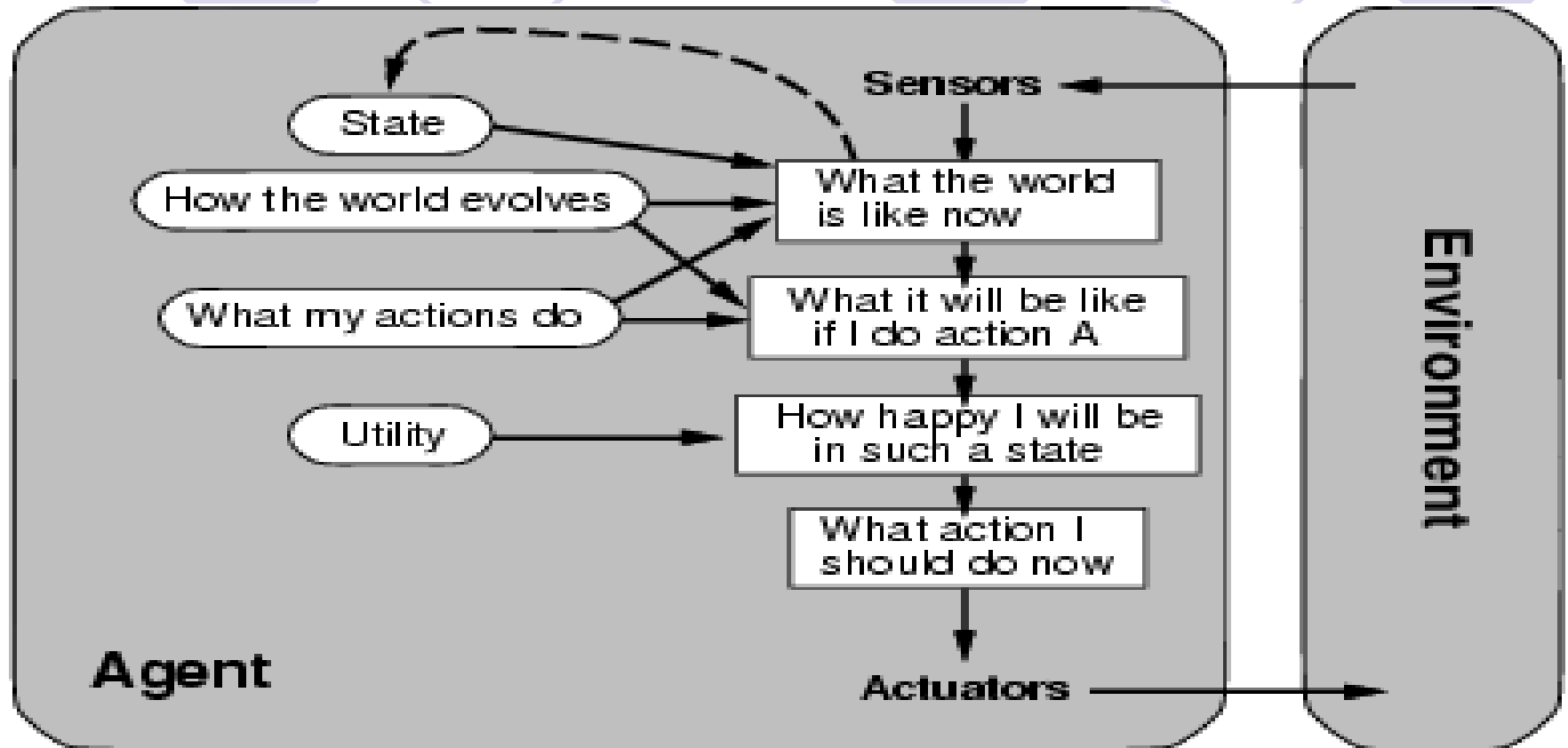
actionSet $\leftarrow$ `POSSIBLE_ACTIONS(state)`

action $\leftarrow$ `ACTION_THAT_LEADS_TO_GOAL(actionSet)`

state $\leftarrow$ `UPDATE_STATE(state, action)`

return action

Utility-based agents



Utility-based agents structure

- Function `UTILITY_BASED_AGENT(PERCEPT)` return action
`static state`, a description of the current world state
`goal`, a description of the goal to achieve may be in terms of state

state $\leftarrow$ *UPDATE_STATE*(*state*, *percept*)

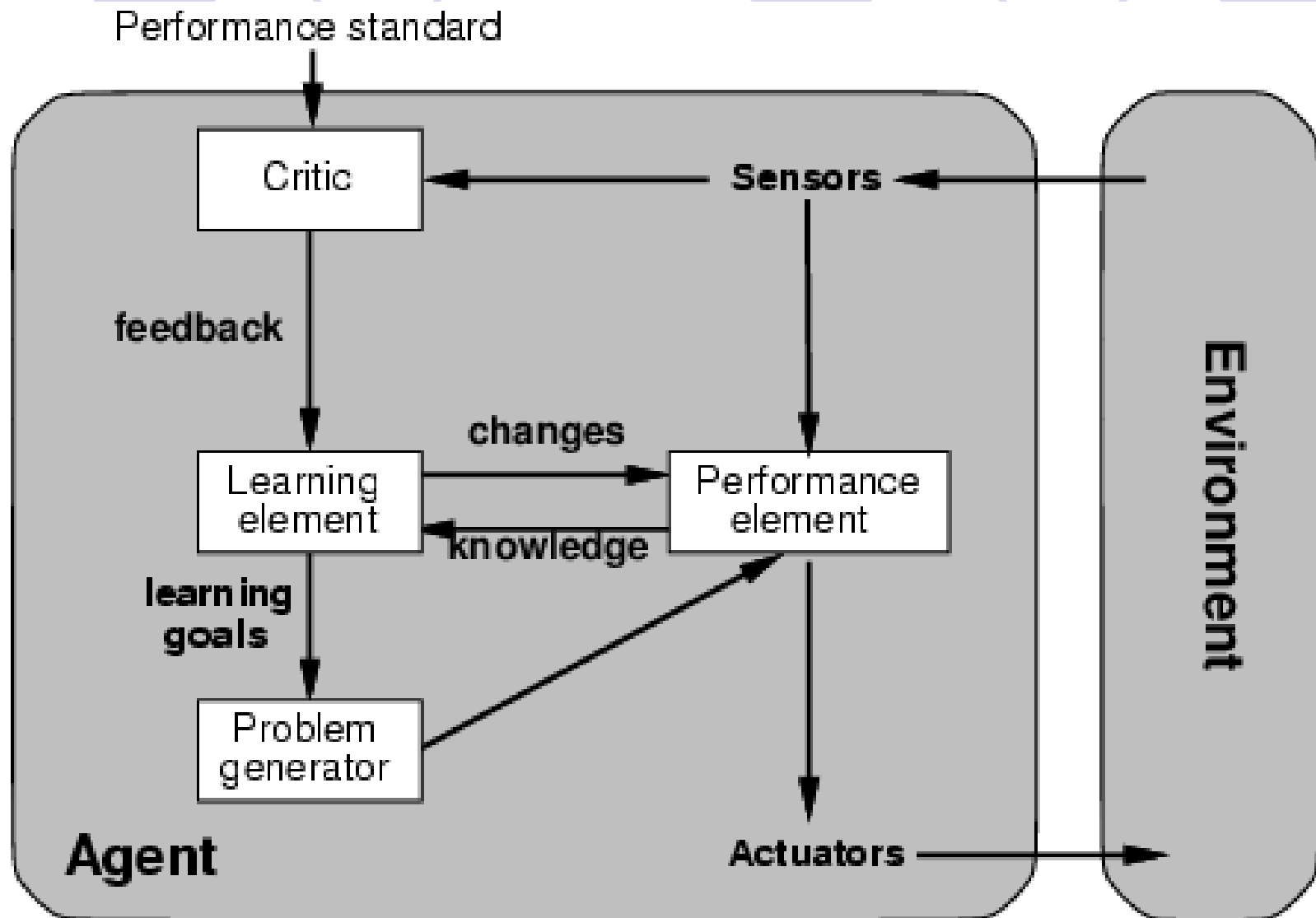
actionSet $\leftarrow$ *POSSIBLE_ACTIONS*(*state*)

action $\leftarrow$ *BEST_ACTION*(*actionSet*)

state $\leftarrow$ *UPDATE_STATE*(*state*, *action*)

return action

Learning agents



Types of Environment

- Based on the portion of the environment observable
 - Fully observable: An agent's sensors give it access to the complete state of the environment at each point in time. (chess vs. driving)
 - Partially observable
 - Fully unobservable
- Based on the effect of the agent action
 - Deterministic : The next state of the environment is completely determined by the current state and the action executed by the agent.
 - Strategic: If the environment is deterministic except for the actions of other agents, then the environment is **strategic**
 - Stochastic or probabilistic

Types of Environment

- **Based on the number agent involved**
 - **Single agent** A single agent operating by itself in an environment.
 - **Multi-agent:** multiple agents are involved in the environment
- **Based on the state, action and percept space pattern**
 - **Discrete:** A limited number of distinct, clearly defined state, percepts and actions.
 - **Continuous:** state, percept and action are consciously changing variables
 - Note: one or more of them can be discrete or continuous

Types of Environment cont ...

- **Based on the effect of time**

- **Static**: The environment is unchanged while an agent is deliberating.
- **Dynamic**: The environment changes while an agent is not deliberating.
- **semi-dynamic**: The environment is **semi-dynamic** if the environment itself does not change with the passage of time but the agent's performance score does

- **Based on loosely dependent sub-objectives**

- **Episodic**: The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.
- **Sequential**: The agent's experience is a single atomic "episodes"

Environment types example

	Chess with a clock		Chess without a clock	Taxi driving
Fully observable	Yes	Yes		No
Deterministic		Strategic	Strategic	No
Episodic		No	No	No
Static		Semi	Yes	No
Discrete	Yes	Yes		No
Single agent		No	No	No

- The environment type largely determines the agent design
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent